

A strict protocol separates model transfer from label availability

a Held-out *C. roseus*

Reference labels; 256 leaf cells

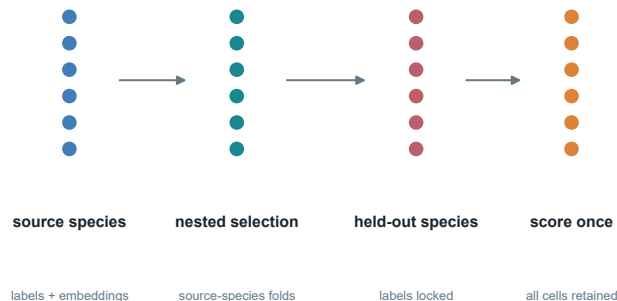


b Nested zero-shot decoding

Target labels are never used in fitting or selection



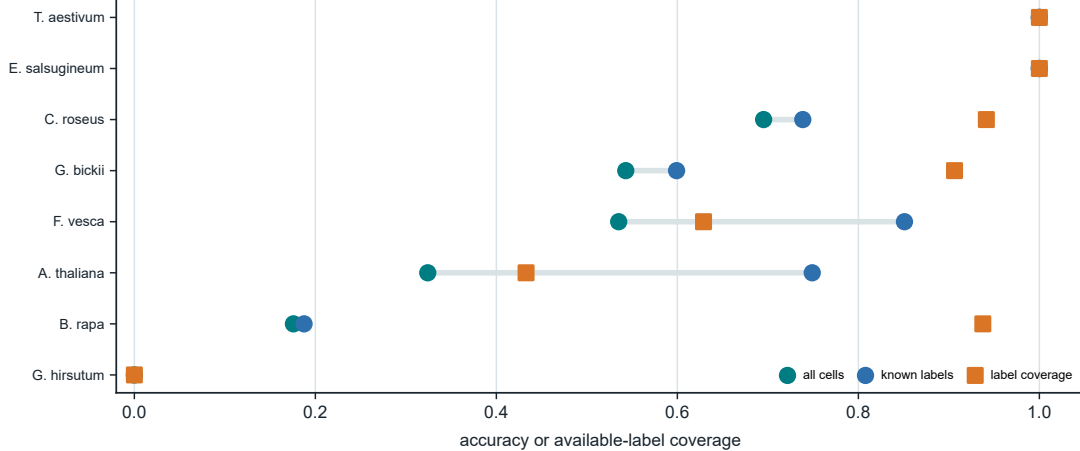
No target cell label enters the decoder, gate choice or calibration.



d

All eight held-out species remain visible in the primary stress test

Teal: all-cell accuracy; blue: conditional accuracy; orange: coverage | fixed-cell bootstrap 95% central interval 0.384-0.415



e

Label-integrity audit

v18 is a companion cohort, not a substituted headline

